VIETNAM NATIONAL UNIVERSITY, HANOI

INTERNATIONAL SCHOOL

INS3064

MULTIMEDIA DESIGN AND WEB DEVELOPMENT

FINAL PROJECT REPORT

CREAT WEBSITE FOR ELECTRONICS SHOP

Group members

Nguyễn Huy Dũng - 23070046

Đỗ Hữu Hiệp - 23070206

Lê Minh Dương - 23070052

Dương Hải Đông - 23070968

SUPERVISOR: PhD. Phạm Đức Thọ

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Finally, we express our sincere appreciation to the Faculty and the University. We are grateful for the exceptional academic environment and the comprehensive resources provided, which served as the essential foundation for our research and the successful execution of this project.

STUDENT DECLARATION

I certify that the assignment submission is entirely my own work and I fully understand the consequences of plagiarism. I declare that the work submitted for assessment has been carried out without assistance other than that which is acceptable according to the rules of the specification. I certify I have clearly referenced any sources and any artificial intelligence (AI) tools used in the work. I understand that making a false declaration is a form of malpractice.

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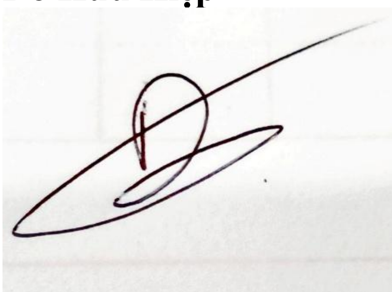
Nguyễn Huy Dũng **Đỗ Hữu Hiệp**



Dương Hải Đông **Lê Minh Dương**

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CHAPTER 1: INTRODUCTION

1.1. Rational for Topic Selection

In the era of the Fourth Industrial Revolution, the transition from traditional shopping models to e-commerce has become an inevitable trend. In the field of technology equipment, consumers now prefer researching technical specifications and comparing prices online rather than visiting physical stores. This shift creates a golden opportunity for businesses to implement digital transformation to reach customers effectively. This project was chosen to build ElectronicsShop, a specialized platform catering to diverse users, from students to tech enthusiasts. Developing this modern web application using Native PHP and MySQL not only meets shopping demands but also provides a comprehensive solution for optimizing inventory and sales management.

1.2. Real-world Challenges and System Solutions

One of the greatest challenges that rudimentary sales systems often face is the loss of data control in complex real-world situations, typically the risk of "overselling" when multiple customers place orders for an item with limited stock at the same time. To address this challenge, the system implements a transaction synchronization solution through a strict data management mechanism at the database level. When a user confirms an order, the system performs a lock on the corresponding product data row, ensuring that all inventory checks and quantity deductions are executed as a single, exclusive block of commands. This technical solution eliminates inconsistencies between the actual physical stock and the orders displayed on the management system.

Another challenge involves personal information security and user session management in a high-risk network environment. The system solves this by applying modern encryption algorithms to protect passwords from the registration stage, ensuring that sensitive customer data is not compromised even in unauthorized access situations. For the login state persistence feature, the project implements an authentication solution using temporary identifiers and secure hashes stored separately. Furthermore, resource management issues, such as product images, are resolved by using unique identifiers for files and an automatic physical data cleanup mechanism when a product is removed, helping the system optimize storage space and maintain stable server performance.

1.3. Importance and Effectiveness of the Solution

This technological solution plays a vital role in creating a transparent and absolutely secure transaction environment for buyers. In the electronic components sector, information accuracy and stock availability are key factors determining a store's reputation. By displaying real-time inventory and implementing an automated deduction process, the system allows customers to fully trust the displayed information, thereby eliminating negative experiences such as successfully placing an order only to be later notified of an abrupt out-of-stock status. The integration of a responsive interface across multiple devices further enables users to look up component information and shop anytime, anywhere, directly on their personal mobile devices with maximum smoothness.

For the management team, this solution is a pivotal tool for enhancing labor productivity and protecting business profits through the comprehensive automation of the process from receiving orders to updating online revenue. The tightly integrated employee authorization system plays a crucial role in protecting sensitive administrative areas, ensuring that only personnel with corresponding authority can interfere with inventory or member data. By systematically capturing precise product quantities and transaction history, managers can develop accurate restocking plans or promotional programs, minimizing long-term inventory stagnation or shortages of core items, thus creating a solid foundation for the sustainable development of the business.

CHAPTER 2: SYSTEM DESIGN

2.1. User Stories: From a Traditional Store to a Digital Management System

The story of ElectronicsShop begins with a traditional technology retailer facing the rapid shift in consumer behavior toward online shopping. As the market evolved, the store owner realized that relying solely on physical foot traffic was no longer sustainable. To survive and thrive, the business required a digital tool that could not only facilitate online sales but also provide a robust framework for managing an expanding inventory and a growing team of employees.

This transition led to the development of specific functional requirements, categorized into the following narratives:

The Business Expansion Story (Admin & Management) As the shop owner plans to scale the business, the primary challenge is maintaining control over a vast array of technical components and managing staff responsibilities. The owner needs a centralized Admin Dashboard to oversee all products and stock levels in real-time to ensure the business never misses a sale due to inventory errors. More importantly, as the team grows, the owner requires a Role Management system to distinguish between general staff and administrators. This ensures that while employees can assist with sales, only trusted managers have the authority to modify sensitive data, such as product pricing or user permissions, thereby protecting the store's integrity.

The Online Sales Evolution (Customer Experience) From the perspective of the modern customer, the transition to an online platform must offer a seamless and professional experience that mirrors the reliability of a physical store. Customers, such as engineering students or tech professionals, need a high-performance Search Engine to filter products by name or SKU, allowing them to find precise components for their work without manual assistance. To facilitate a smooth transaction, the system must provide an Intelligent Shopping Cart and a Secure Checkout process that guarantees their purchase by locking inventory the moment a transaction begins. Furthermore, to build long-term loyalty, the platform includes a Persistent Login feature, enabling customers to return and shop frequently without the friction of repeated authentication.

Ultimately, the birth of the ElectronicsShop web application is the direct answer to this merchant's need for a scalable, secure, and automated environment that bridges the gap between traditional retail and the modern digital marketplace.

2.2. Site map

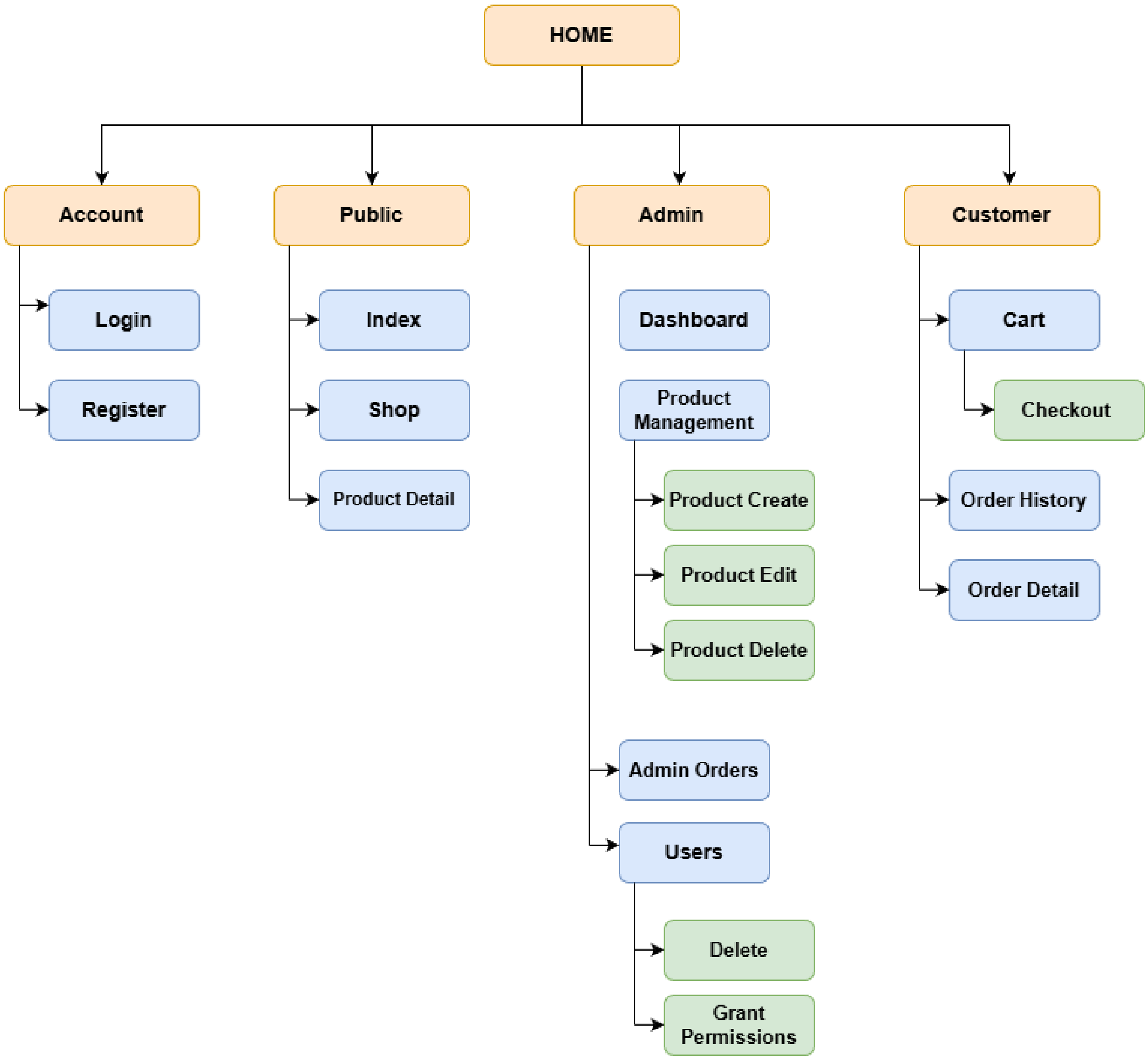

Figure 2.1: Site map of ElectronicsShop

The ElectronicsShop system's sitemap is designed with a clear functional hierarchy, optimizing the user experience for both customers and administrators. From the homepage, the system branches into four main sections: Account for user authentication, Public containing public product pages, Admin focusing on advanced management tools, and Customer serving the individual shopping process. Specifically, this structure demonstrates strict decentralization through low-level data processing branches (green) such as adding, editing, deleting products and granting user permissions, ensuring security and stable operation for the entire e-commerce platform.

2.3. Entity Relationship Diagram

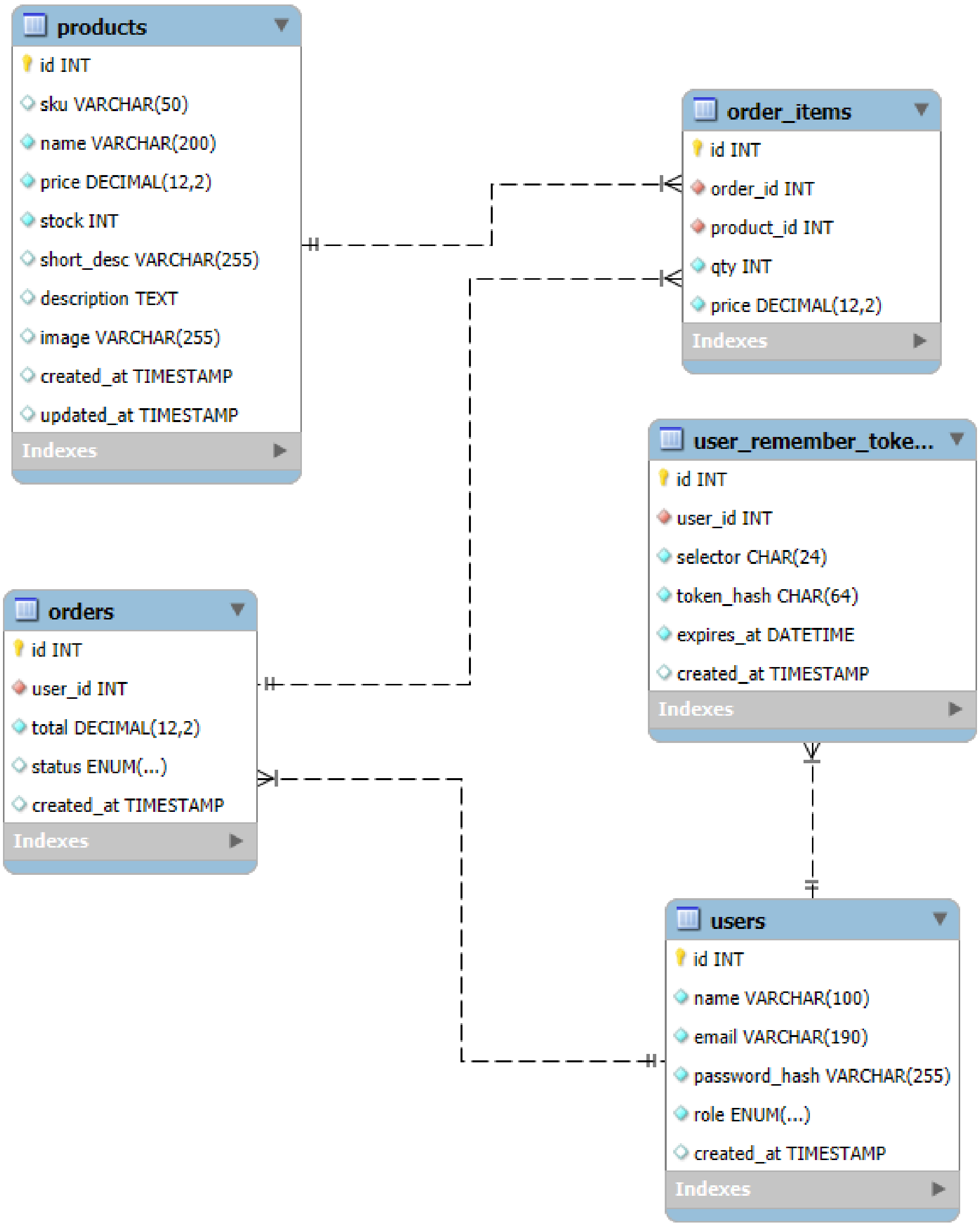

Figure 2.2. Entity-Relationship Diagram

The database system is designed using a relational model to optimize user management and the online sales process. At the core of the system is the *users* entity, which stores identification details, hashed passwords, and user roles for access control. To support secure "remember me" functionality, the *user_remember_tokens* entity is linked to the users via a one-to-many (1:N) relationship, managing authentication tokens and session expiration. For business operations, the *products* entity stores comprehensive item details, including SKU, name, unit price, and stock levels. The transaction workflow is managed through the *orders* entity, which records the total value and status of each purchase associated with a specific user. Finally, the *order_items* entity serves as a critical bridge in the many-to-many (N:N) relationship between orders and products; it not only links these two entities but also captures transaction-specific data such as quantity and the actual price at the time of purchase, ensuring data integrity across the entire system.

Physical database design:

Attribute	Data Type	Width	Meaning
id	INT	10	Primary Key (Auto-increment)
name	VARCHAR	100	Full name of the user
email	VARCHAR	190	Unique email address (Login)
password_hash	VARCHAR	255	Hashed password
role	ENUM		User role (admin, user)
created_at	TIMESTAMP		Account creation timestamp

Table 2.1. Users Table

Attribute	Data Type	Width	Meaning
id	INT	10	Primary Key
user_id	INT	10	Foreign Key referencing Users table

selector	CHAR	24	Unique token identifier
token_hash	CHAR	64	Hashed validator token
expires_at	DATETIME		Token expiration date
created_at	TIMESTAMP		Token creation timestamp

Table 2.2. User Remember Tokens Table

Attribute	Data Type	Width	Meaning
id	INT	10	Primary Key
sku	VARCHAR	50	Stock Keeping Unit (Unique)
price	DECIMAL	12,2	Unit price
stock	INT	11	Inventory quantity
short_desc	VARCHAR	255	Brief product summary
description	TEXT		Full product description
image	VARCHAR	255	Product image file path
created_at	TIMESTAMP		Date product was added
updated_at	TIMESTAMP		Last modification date

Table 2.3. Products Table

Attribute	Data Type	Width	Meaning
id	INT	10	Primary Key
user_id	INT	10	Foreign Key referencing Users table
total	DECIMAL	12,2	Total order amount
status	ENUM	255	Order status (pending, paid, shipping, etc.)
created_at	TIMESTAMP		Date the order was placed

Table 2.4. Orders Table

Attribute	Data Type	Width	Meaning
id	INT	10	Primary Key
order_id	INT	10	Foreign Key referencing Orders table
product_id	INT	10	Foreign Key referencing Products table
qty	INT	11	Quantity purchased
price	DECIMAL	12,2	Price at the time of purchase

Table 2.5. Order Items Table

CHAPTER 3: IMPLEMENTATION

3.1. Sample Source Code

The ElectronicsShop website's source code is built on a pure PHP architecture combined with a MySQL database, creating a synchronized content management and e-commerce platform. The entire source code is organized in a modular fashion, clearly separating files for business logic, session management, and the display interface, ensuring smooth operation and easy maintenance. The core functions of the website include multi-level product catalog management, a secure online ordering process, and a dashboard management system for store owners. A key highlight of this project is the integration of groundbreaking features such as automated inventory management using database transactions, an intelligent personnel authorization system, and automatic resource cleanup. These strengths not only help the store solve the challenge of transitioning to an online model but also provide a professional human resource management tool, ready for future business expansion.

3.1.1. Secure Transaction and Inventory Integrity Feature

The most critical feature of the system is the secure transaction management mechanism implemented in the online checkout process within the *checkout.php* file. In an online retail environment, ensuring accurate inventory levels is a major challenge, especially when multiple users access and purchase a single technical component simultaneously. This feature is directly applied to prevent "overselling" and ensures that order information is only recorded when all stock verification and deduction steps are successfully completed. When a customer confirms an order, the system automatically triggers a private transaction session, temporarily isolating these data changes from other queries until final confirmation. The effectiveness of this feature results in absolute customer trust in the system while helping the owner avoid risks of inventory discrepancies and revenue loss due to data synchronization errors.

Sample Code Implementation:

```php
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
  try {
    $pdo->beginTransaction();

    $ids = array_keys($cart);
    $in = implode(',', array_fill(0, count($ids), '?'));
    // Khóa dòng để tránh xung đột kho
    $st = $pdo->prepare("SELECT * FROM products WHERE id IN ($in) FOR UPDATE");
    $st->execute($ids);
    $products = $st->fetchAll();

    $map = [];
    foreach ($products as $p) $map[(int)$p['id']] = $p;

    $total = 0.0;
    foreach ($cart as $pid => $qty) {
      if (!isset($map[$pid])) throw new Exception("Sản phẩm ID $pid không tồn tại.");
      $stock = (int)$map[$pid]['stock'];
      if ($stock < $qty) throw new Exception("Sản phẩm '{$map[$pid]['name']}' không đủ hàng (còn $stock).");
      $total += (float)$map[$pid]['price'] * $qty;
    }
```

```php
// Tạo đơn
$ins = $pdo->prepare("INSERT INTO orders(user_id, total, status) VALUES(?, ?, 'paid')");
$ins->execute([current_user()['id'], $total]);
$orderId = $pdo->lastInsertId();
```

```php
// Tạo chi tiết + Trừ kho
$insItem = $pdo->prepare("INSERT INTO order_items(order_id, product_id, qty, price) VALUES(?,?,?,?)");
$updStock = $pdo->prepare("UPDATE products SET stock = stock - ? WHERE id=?");

foreach ($cart as $pid => $qty) {
  $price = (float)$map[$pid]['price'];
  $insItem->execute([$orderId, $pid, $qty, $price]);
  $updStock->execute([$qty, $pid]);
}
```

```php
$pdo->commit();
$_SESSION['cart'] = []; // Xóa giỏ

header("Location: checkout.php?success=1&order=" . $orderId);
exit;

catch (Throwable $e) {
$pdo->rollBack();
$err = "Lỗi thanh toán: " . $e->getMessage();
```

Source Code Analysis: The code operates through three pivotal stages: transaction initialization, lock execution, and confirmation or cancellation. The greatest advantage lies in the SELECT ... FOR UPDATE statement, which is an advanced row-level locking technique in MySQL. When this command is executed, the system prevents any other process from interfering with the quantity of that product until the *commit()* command is called. Sophistication is also found in the *try...catch* block, where the *rollBack()* command acts as a "safety brake," reverting all data to its original state if any product in the cart has insufficient quantity or a system error occurs. This mechanism ensures "Atomicity" - either all products in the order are successfully processed, or no inventory changes are made at all.

3.1.2. Multi-tier Personnel Authorization and Access Security Feature

The system integrates a flexible staff management and access authorization feature, designed to address administrative needs as the store expands its team. This feature utilizes a Role-Based Access Control (RBAC) mechanism to determine the specific permissions of each individual upon logging into the system via the *auth.php* file. The shop owner uses this feature to segregate duties between staff and customers, ensuring that the technical team can manage products and orders without interfering with high-level system configurations. The result is a professional operational workflow that minimizes the risk of sensitive information leakage and allows the owner to easily monitor online staff activities through the central dashboard.

Source Code Implementation (Authentication & Authorization):

```
function current_user(): ?array {
  return $_SESSION['user'] ?? null;
}
```

```
function require_login(): void {
  if (!current_user()) {
    header("Location: /login.php");
    exit;
  }
}
```

```php
if (!current_user()) {
  try {
    // Biến $pdo được gọi từ file bên ngoài (global scope)
    global $pdo;
    if (isset($pdo) && $pdo instanceof PDO) {
      remember_me_try_login($pdo);
    }
  } catch (Throwable $e) {
    // bỏ qua lỗi
  }
}
```

```php
function require_admin(): void {
  require_login();
  if ((current_user()['role'] ?? '') !== 'admin') {
    http_response_code(403);
    exit("Bạn không có quyền truy cập trang này.");
  }
}
```

Source Code Implementation (Admin Management Actions):

```php
// --- CASE 1: XÓA USER ---
if ($action === 'delete') {
  // 1. Bảo vệ: Không cho tự xóa chính mình
  if ($uid === $currentUserId) {
    $_SESSION['flash'] = "Lỗi: Bạn không thể tự xóa tài khoản của mình.";
  } else {
    try {
      // 2. Thực hiện xóa
      $st = $pdo->prepare("DELETE FROM users WHERE id=?");
      $st->execute([$uid]);
      $_SESSION['flash'] = "Đã xóa thành công User #$uid.";
    } catch (PDOException $e) {
      // 3. Xử lý lỗi nếu User đã có đơn hàng (Ràng buộc khóa ngoại)
      if ($e->getCode() == '23000') {
        $_SESSION['flash'] = "Không thể xóa: User này đã có dữ liệu đơn hàng cũ.";
      } else {
        $_SESSION['flash'] = "Lỗi Database: " . $e->getMessage();
      }
    }
  }
  header("Location: users.php"); exit;
}
```

```php
// --- CASE 2: CẬP NHẬT QUYỀN (Code cũ của bạn) ---
if ($action === 'update') {
    $role = $_POST['role'] ?? 'user';
    if (!in_array($role, ['user','admin'], true)) $role = 'user';

    // Không cho tự hạ quyền chính mình
    if ($uid === $currentUserId && $role !== 'admin') {
      $_SESSION['flash'] = "Không thể tự hạ quyền chính mình.";
      header("Location: users.php"); exit;
    }

    $st = $pdo->prepare("UPDATE users SET role=? WHERE id=?");
    $st->execute([$role, $uid]);

    $_SESSION['flash'] = "Đã cập nhật quyền.";
    header("Location: users.php"); exit;
}
```

Source Code Analysis: The operation of this code is based on validating the user's session state immediately upon accessing administrative pages. The advantage lies in its centralized and "secure by default" nature; by defining the validation function in a single *auth.php* file, this security layer can be embedded into numerous administrative files such as *dashboard.php* or *users.php* with just one line of code. The code follows the "Default Deny" principle, meaning if the dual conditions of being logged in and possessing an admin role are not simultaneously met, the system immediately terminates the process using the *exit()* command. This creates a robust security boundary, preventing any attempts to tamper with component inventory data or order information by regular users or unauthorized external access.

3.1.3. User Order Status Tracking and Administrative Update Feature

The system provides a closed-loop process that connects information between administrators and customers through real-time order status updates. This feature allows the Admin to control the store's operational flow while enabling users to actively track the journey of the components they have ordered, from the pending stage to completion. In an e-commerce environment, this is a key feature for building trust and transparency in transactions between buyers and sellers.

Technical Analysis and Strengths:

- Centralized Administration: At the *admin/orders.php* file, administrators can change the status of any order (Pending, Paid, Shipping, Completed, Cancelled) through an intuitive interface. Every change is executed using the POST method and Prepared Statements to ensure data security and prevent unauthorized access from altering transaction states.
- Visual Feedback for Users: When customers access *order_history.php*, they see an order list categorized by colors corresponding to the status (e.g., green for paid, orange for pending). The system uses the *strtotime()* function to format the order date correctly according to Vietnam time, helping users easily reconcile their shopping history.

Source Code Implementation (Admin Order Management):

```php
// Xử lý cập nhật trạng thái
if ($_SERVER['REQUEST_METHOD'] === 'POST' && isset($_POST['action']) && $_POST['action'] === 'update_status') {
    $oid = (int)$_POST['order_id'];
    $st  = $_POST['status'];
    // Cập nhật
    $up = $pdo->prepare("UPDATE orders SET status = ? WHERE id = ?");
    $up->execute([$st, $oid]);
    $_SESSION['flash'] = "Cập nhật trạng thái đơn #$oid thành công.";
    header("Location: orders.php"); exit;
}
```

```php
// Lấy tất cả đơn hàng (Kèm tên người mua)
$sql = "SELECT o.*, u.name as user_name, u.email as user_email
        FROM orders o
        JOIN users u ON o.user_id = u.id
        ORDER BY o.created_at DESC";
$orders = $pdo->query($sql)->fetchAll();
?>
```

Source Code Implementation (User Order Detail Security):

```php
$orderId = (int)($_GET['id'] ?? 0);
$currentUser = current_user();

$st = $pdo->prepare("SELECT * FROM orders WHERE id = ?");
$st->execute([$orderId]);
$order = $st->fetch();
```

```php
if ($currentUser['role'] !== 'admin' && (int)$order['user_id'] !== (int)$currentUser['id']) {
    echo "<div class='alert error'>Bạn không có quyền xem đơn hàng này.</div>";
    require_once __DIR__ . "/includes/footer.php";
    exit;
}
```

Source Code Analysis: The functionality of this feature relies on data synchronization between the *orders* table and the user interface. The advantage lies in the processing logic at *admin/orders.php*: after executing the UPDATE command, the system utilizes the Post-Redirect-Get (PRG) Pattern to redirect the page, preventing data from being resubmitted if the Admin refreshes the page. On the user side, using conditional variables to change the CSS color codes (*$color*) makes information extremely easy to understand and visually appealing, subtly enhancing the User Experience (UX).

3.2. Images of final Applications

YouTube URL: https://youtu.be/z2C3125DxqI

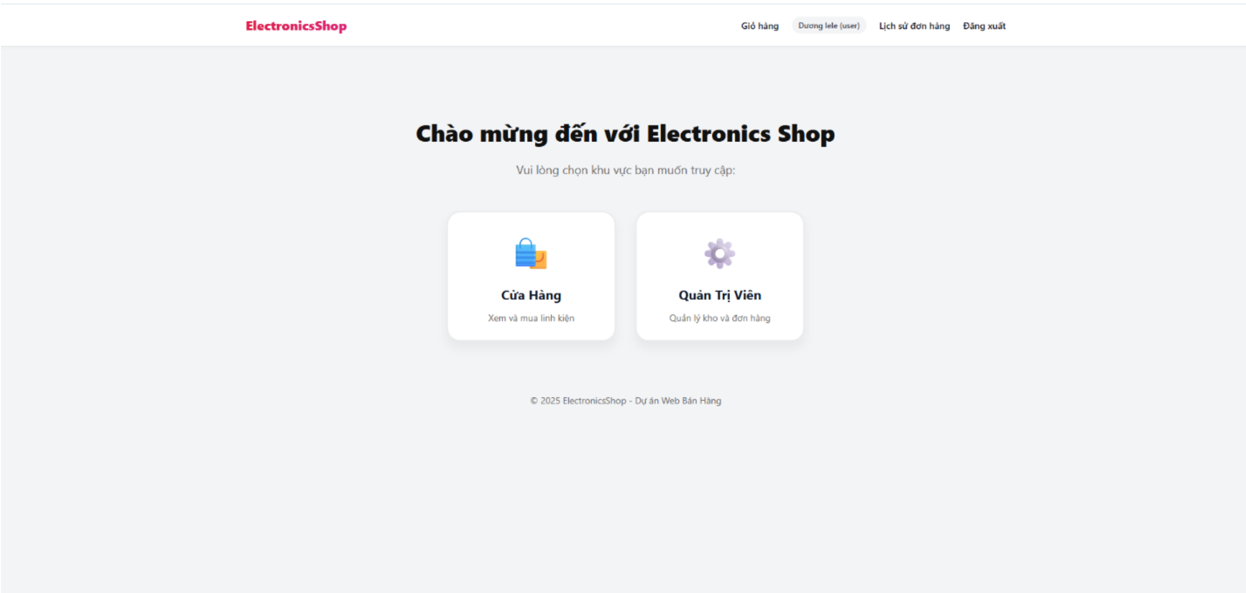

Figure 3.1. Homepage interface of the ElectronicsShop system

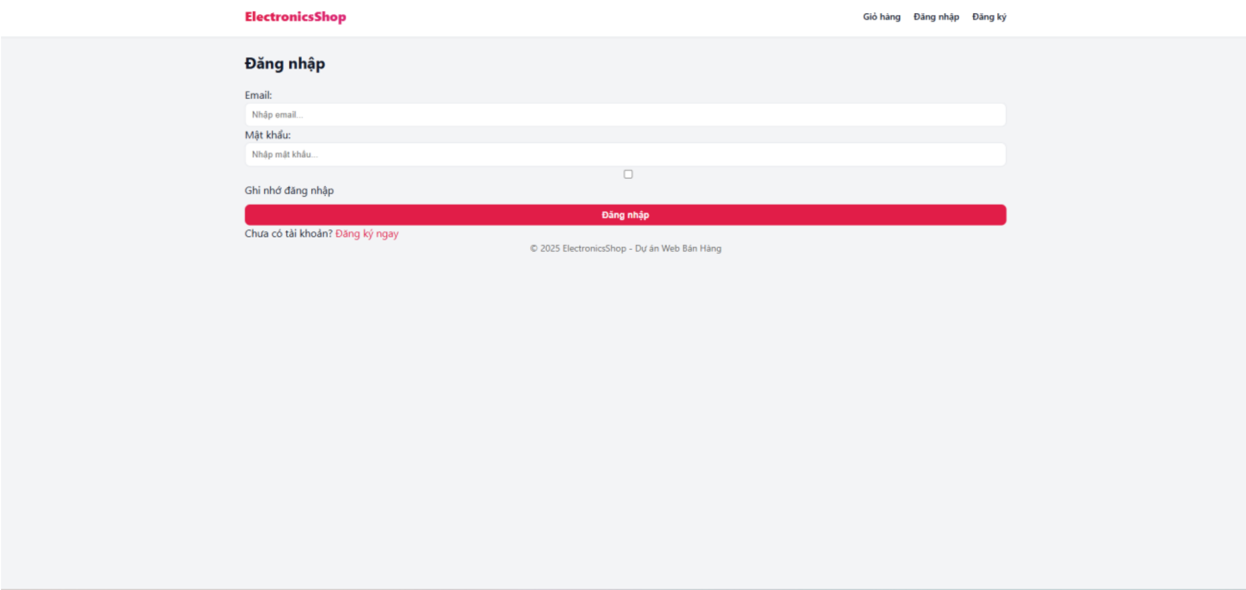

Figure 3.2. Login page

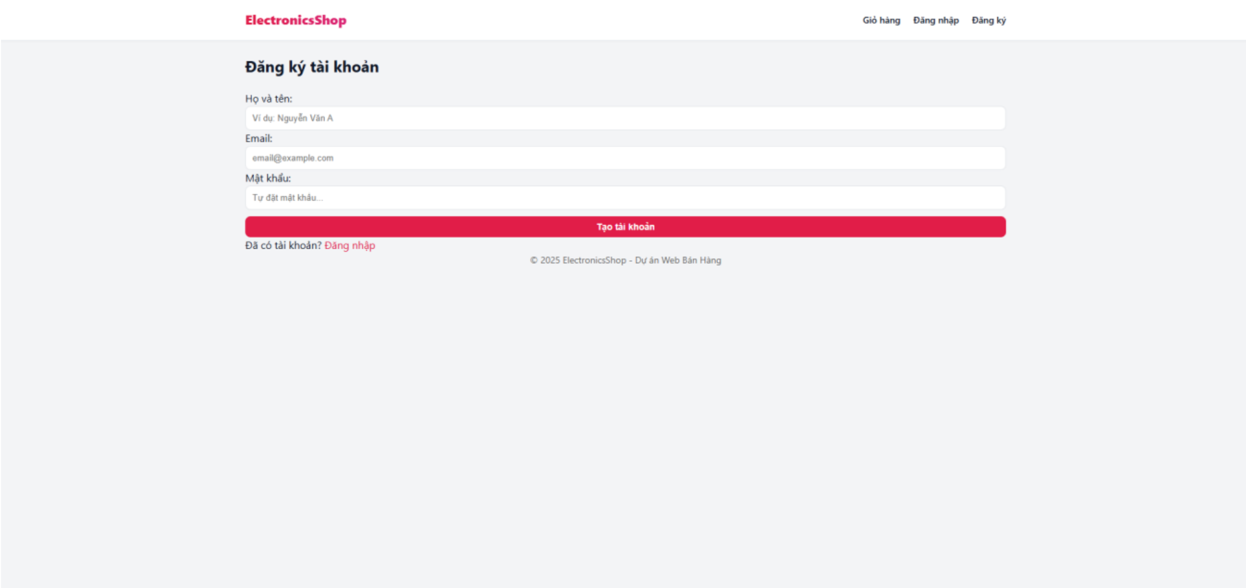

Figure 3.3 Account registration page

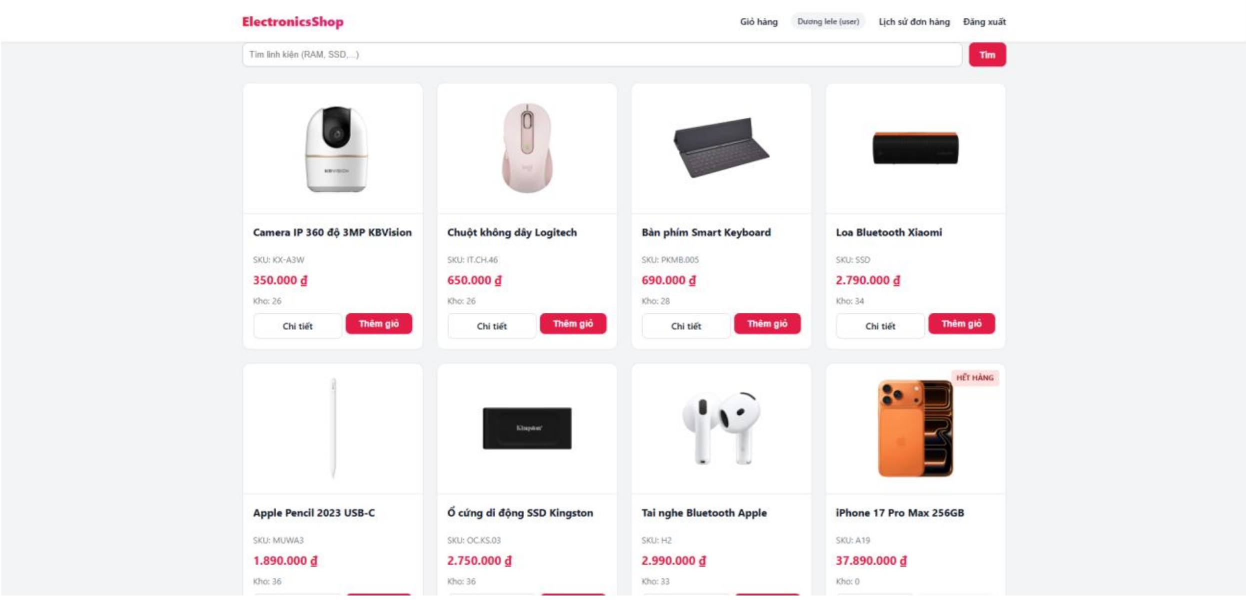

Figure 3.4. Shopping interface

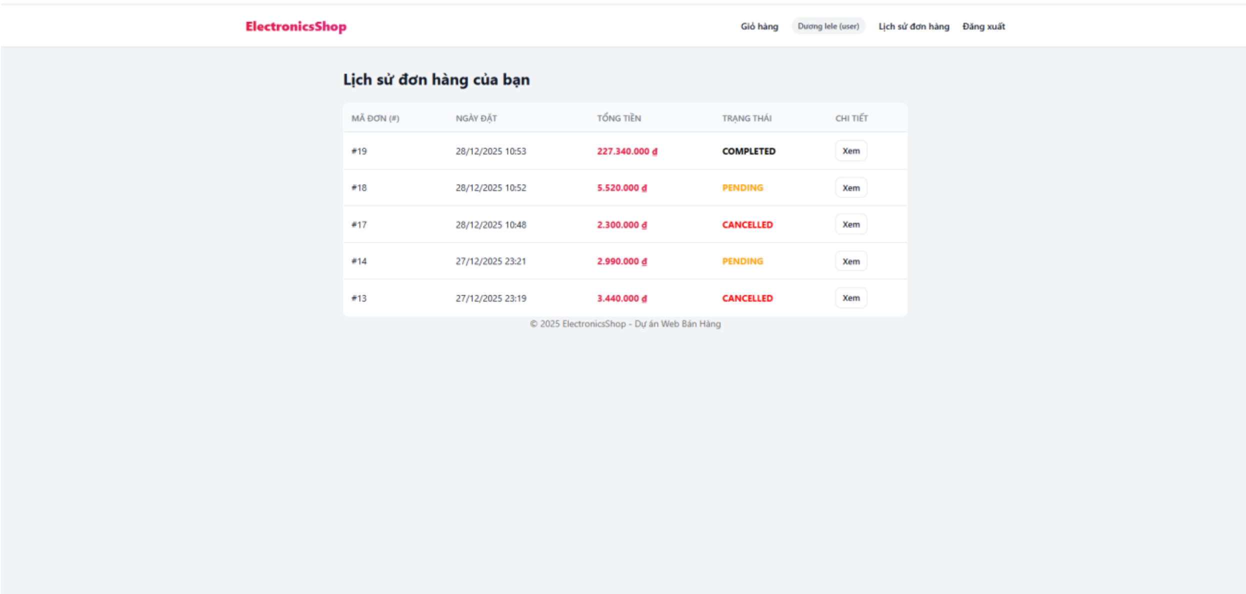

Figure 3.5. Purchase history management interface.

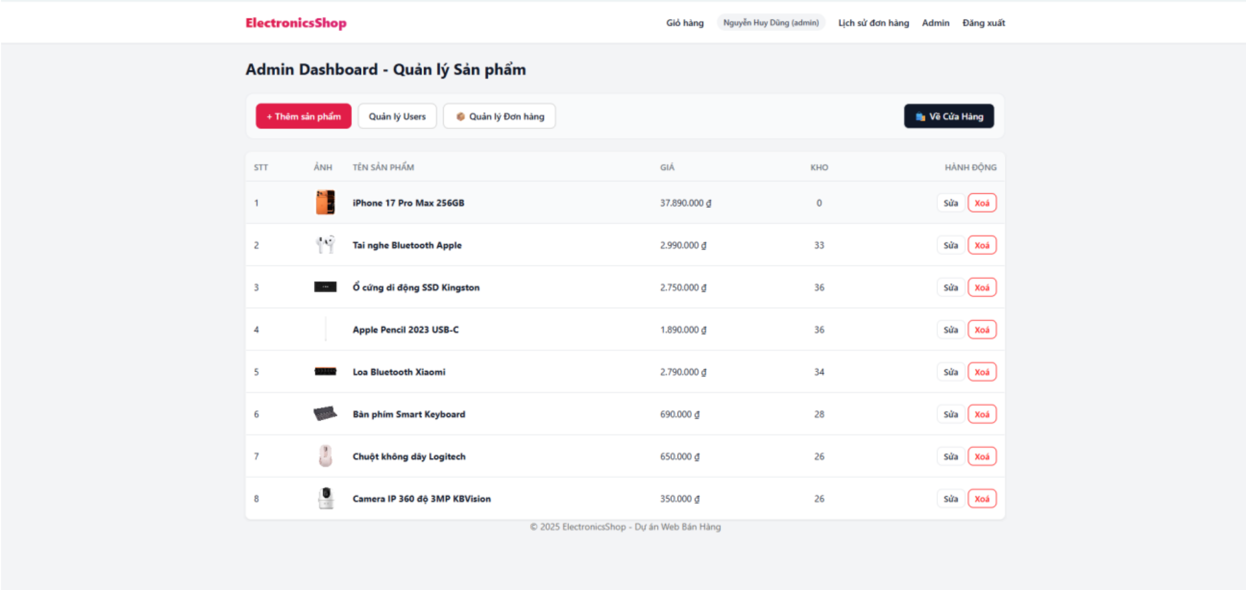

Figure 3.6. Admin's product management page

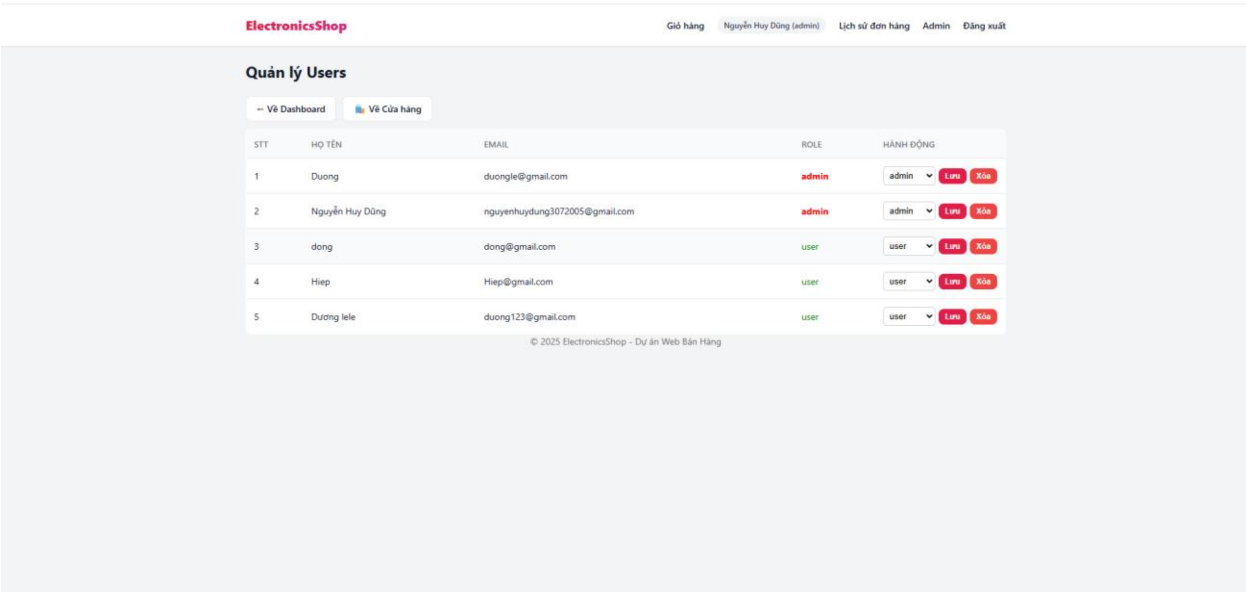

Figure 3.7. Admin's Users account management page.

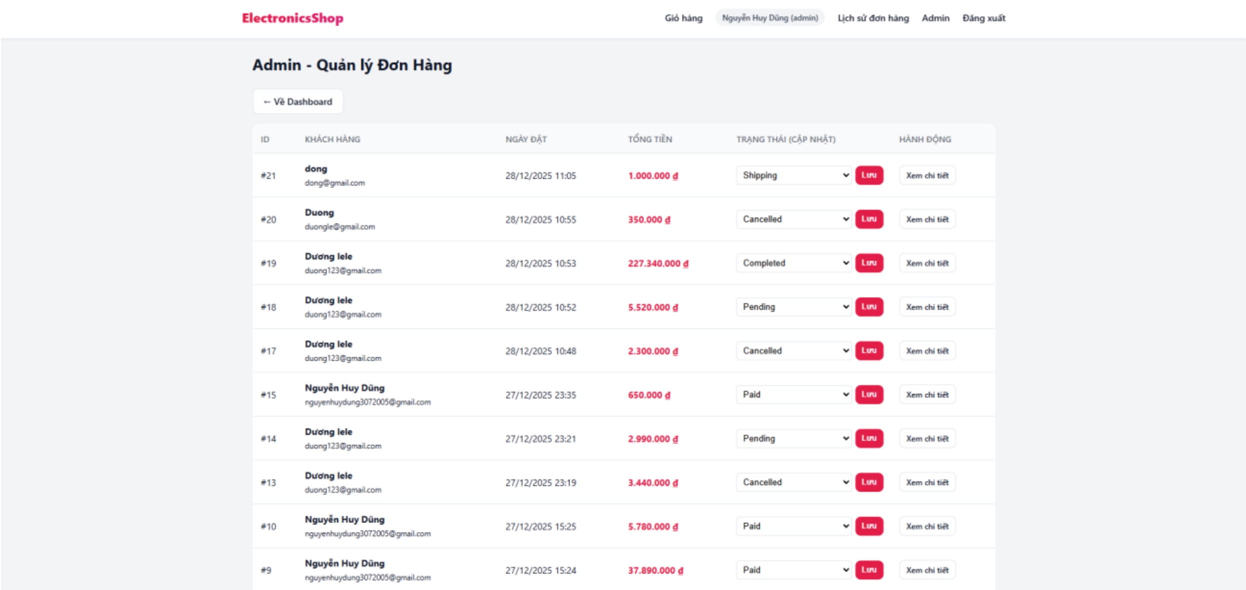

Figure 3.8. Admin's order management page

3.3. GitHub Repository evidences

GitHub link: https://github.com/duonglele/final-project1/commits/main/

GitHub commit history:

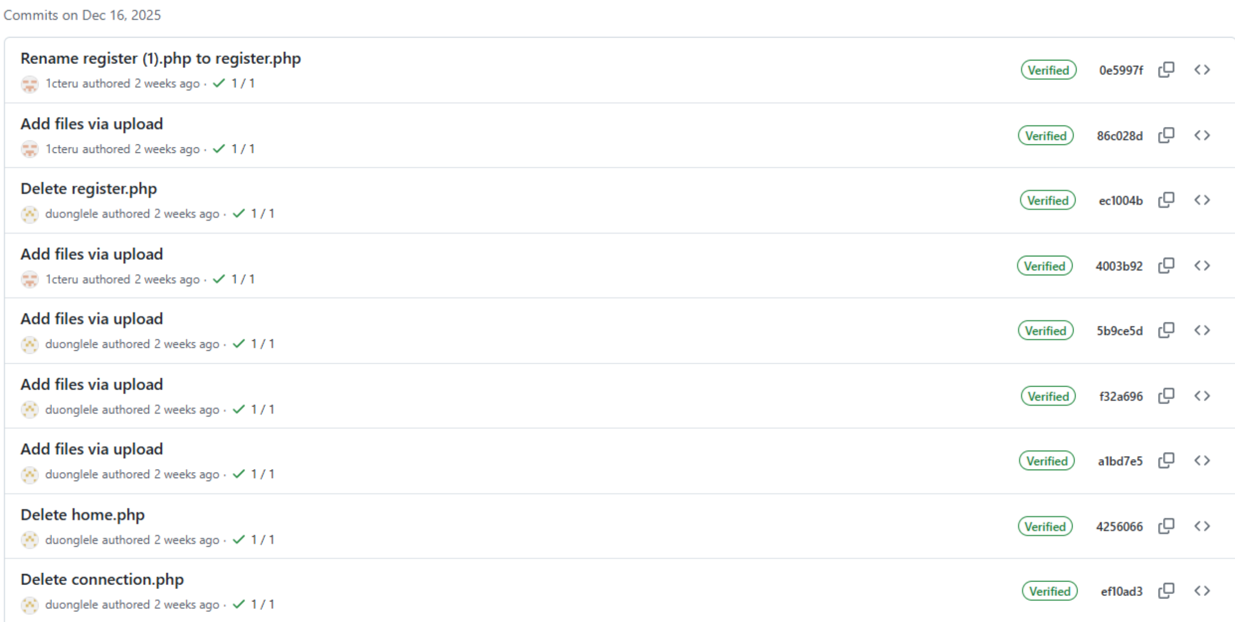

Figure 3.9. GitHub history

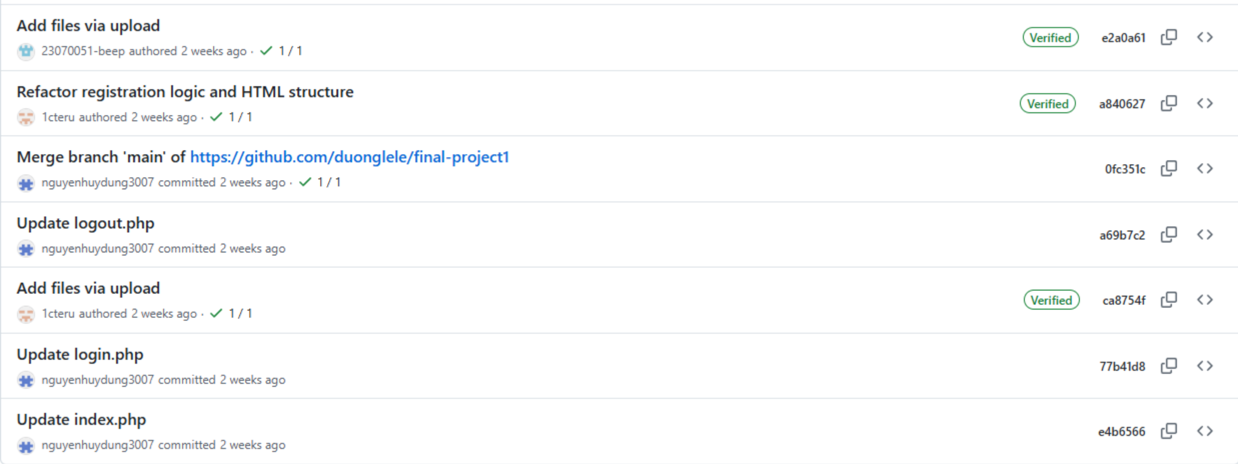

Figure 3.10. GitHub history

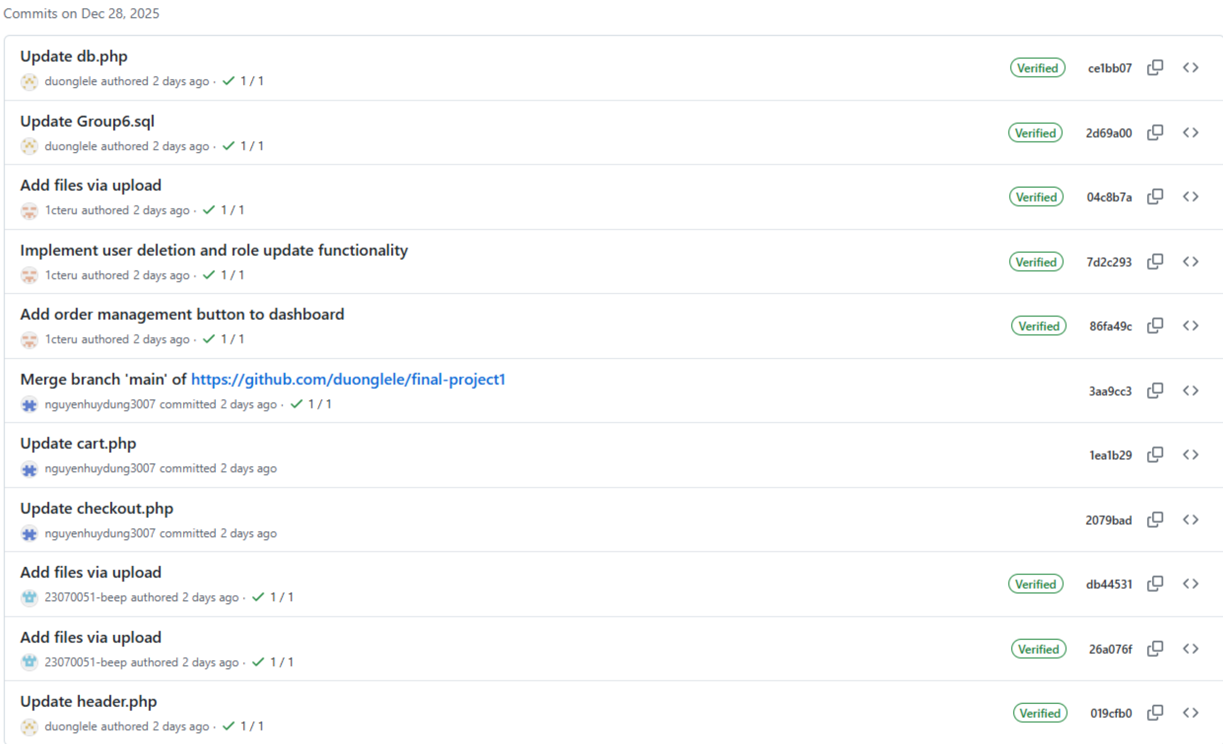

Figure 3.11. GitHub history

CHAPTER 4: CONCLUSION

4.1. What went well

The project successfully achieved the goal of building a stable and practical electronic component e-commerce system, effectively connecting a modern user interface with a database on a real-world InfinityFree hosting environment. One of the most significant functional successes our team achieved was establishing a transparent and rigorous order control process for both buyers and sellers. For customers, the system provides an intuitive order history page, allowing them to track the journey of their products from the moment of placement to completion, which significantly enhances trust in the store's service. Conversely, our team equipped administrators with a centralized dashboard to update order statuses in real-time, ensuring that every change in payment or shipping status is instantly synchronized across the entire system. Furthermore, the implementation of a secure transaction mechanism in the checkout processing file ensured absolute integrity for the inventory, eliminating data discrepancies during simultaneous transactions. Finally, the personnel authorization system through a central security layer helped completely segregate permissions, creating a professional and secure management environment for the shop owner.

4.2 What did not go well

Despite achieving many positive results, the system still faced certain limitations arising from both subjective and objective factors during implementation. Due to the nature of deploying on the InfinityFree free hosting environment, the system occasionally experienced slow response times and time zone discrepancies between the PHP server and MySQL, forcing our team to manually intervene with time configuration commands in the source code to ensure invoice accuracy. Additionally, the website's payment process is currently limited to a simulation of Cash on Delivery (COD), as we have not yet deeply integrated automated online payment gateways such as VNPay or MoMo to fully automate the cash flow. Another shortcoming to mention is the automated feedback system, as the website still lacks the ability to send immediate confirmation emails to customers or low-stock alerts to administrators, which reduces the proactiveness of the store's operational management.

4.3 Lessons learned and further improvements

The process of researching and implementing the ElectronicsShop project provided our team with valuable lessons in system programming thinking and collective teamwork skills to create a complete product. Through solving technical problems together, the team better understood the importance of using modern security techniques such as password hashing and parameterized queries to protect user information against common cyberattacks. Specifically, the experience of handling data contention through row-level locking commands in the database helped the team members learn how to maintain system stability in environments with many simultaneous users. Our team also practiced the ability to manage shared source code and optimize business logic to ensure the final product achieved the highest level of consistency and efficiency.

Looking ahead, our team's future development for the project will focus on completing automated payment modules to create a more seamless and professional shopping experience for customers. Simultaneously, the system's administrative site will be upgraded with advanced reporting tools in the form of visual charts, helping the shop owner easily monitor revenue and shopping trends directly from the central dashboard. Finally, integrating automated notification services via Email or Telegram will be a top priority to improve the interaction speed between the system and the manager, ensuring that every fluctuation in orders and inventory is handled promptly to enhance the service quality of ElectronicsShop.